



Are extreme returns priced in the stock market? European evidence



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ABSTRACT

This paper revisits some recently found evidence in the literature on the cross-section of stock returns for a carefully constructed dataset of euro area stocks. First, we confirm recent results for US data and find evidence of a negative cross-sectional relation between extreme positive returns and average returns after controlling for characteristics such as momentum, book-to-market, size, liquidity and short term return reversal. We argue that this is the case because these stocks have lottery-like characteristics, which is attractive to certain investors. Also, these stocks tend to be very volatile so that arbitrageurs are discouraged from correcting potential mispricing. As a consequence, these stocks are often overpriced and hence face lower expected returns. Second, when we control for extreme returns, the recently found negative relationship between idiosyncratic risk and future returns is less robust. In our models, after adding maximum returns, the relationship is insignificant and sometimes even positive. We also find that idiosyncratic skewness and coskewness play an important role for asset pricing, as predicted by several theoretical models.

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1. Introduction

Despite decades of research, it is still not completely clear what determines cross-sectional variation in expected stock returns. It is well accepted that the four factor model of [Fama and French \(1993\)](#) and [Carhart \(1997\)](#) goes a long way in capturing this return variation. Nevertheless, empirical studies continue to document other characteristics that are related to average stock returns (see [Fama and French \(2008\)](#) for a recent overview). As most of these characteristics do not follow directly from theory, they are subject to the data-mining critique implying the patterns are mere statistical flukes ([Lo and MacKinlay, 1990](#)). In any dataset, some results are bound to be statistically significant just by chance. One way to address this critique is to look for corroborating evidence on other markets or from other periods.

In this paper, we verify the US results of [Bali et al. \(2011\)](#) on a carefully constructed euro area stock market database covering more than 30 years. More specifically, [Bali et al. \(2011\)](#) find a statistically and economically significantly negative relation between

the maximum daily return over the past 1 month and expected stock returns. They argue that this captures individual investors' preference for lottery-like stocks, i.e. stocks that have a low probability of a huge profit and a large probability of a small loss as shown by [Kumar \(2009\)](#). Although this is an idiosyncratic characteristic, demand by individual investors may lead to higher prices (and lower expected returns) for these stocks, given that these investors typically hold underdiversified portfolios, see e.g. [Odean \(1999\)](#) and [Goetzmann and Kumar \(2008\)](#). We show that this effect also exists in the euro area. Moreover, it is unlikely to be arbitrated away by large investors as the typical stock with extreme positive returns is relatively small, illiquid and has relatively high idiosyncratic volatility. Even if short selling these stocks were possible, it would expose arbitrageurs to considerable risk. Hence, it is plausible that individual investors drive the pricing of such stocks.

We also look at idiosyncratic skewness and coskewness and find that the maximum return effect is robust to including total skewness, idiosyncratic skewness or coskewness. Moreover, we find that both idiosyncratic skewness and coskewness are significantly negatively related to expected returns. Lastly, the comovement between maximum returns and idiosyncratic volatility is investigated. Taken on its own, idiosyncratic volatility is negatively related to expected returns. The relation is statistically significant and goes in the direction of the puzzling results of [Ang et al. \(2006\)](#) for the US and [Ang et al. \(2009\)](#) for an international sample of stocks. However, when we control for extreme positive returns,

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the relation between idiosyncratic risk and expected returns becomes less robust. This may also be due to multicollinearity as discussed by Ahn et al. (2013). Nevertheless, correcting for multicollinearity does not change the conclusion that extreme positive returns are negatively priced.

The remainder of the paper is structured as follows. In Section 2, we discuss sample selection, construction of variables and filters. Next, we discuss the main results in Section 3. Section 4 provides the results of a battery of robustness checks and finally, we conclude.

2. Data

2.1. Sample selection

Our sample comprises thirteen European countries: Austria, Belgium, Finland, France, Germany, Greece, Ireland, Italy, Luxembourg, the Netherlands, Portugal, Slovakia and Spain. Following Schmidt et al. (2011) we exclude the other four countries of the euro area (Malta, Estonia, Cyprus and Slovenia) as their financial impact on the euro area is negligible. In 2009, they contributed only 0.80% of the euro area GDP. We are therefore convinced that the current sample is representative for the entire euro area. All data come from Thomson DataStream (TDS) as in Ang et al. (2009). We include both active shares and delisted shares³ thereby minimizing survival bias in our sample. The resulting dataset is then subjected to extensive filtering as described in Appendix A in order to select only common stock issues. This results in a sample of 7861 European companies. For these companies, we download the end-of-month return index (including dividends), unadjusted stock price, market capitalization (MC) and book-to-market ratio (B/M), from 31 December 1979 to 30 June 2011. Additionally, we download the daily total return index and MC over the same period. TDS automatically calculates B/M by dividing the book value per share by market value per share at time t , where book value per share is from the company's last fiscal year (Worldscope item 05476). For the pre-1999 period all data are converted by TDS into synthetic euro. As the risk-free rate, R_f , the monthly money market rate as reported by Frankfurt banks is used.⁴

2.2. Construction of variables

All returns are calculated using the TDS total return indices. Two procedures, inspired by Ince and Porter (2006) and Schmidt et al. (2011), are used to correct errors that occasionally occur in the TDS database. First, we need to tackle decimal errors. Suppose the return index is 101.52 on a particular day and does not change the next day. When TDS correctly stores 101.52 the first day, but erroneously stores it as 1015.20 the second day, using a computed return of 900% instead of the true zero return would obviously distort results. We call this a right-decimal error, because the decimal moved erroneously to the right. Alternatively, a left-decimal error occurs when TDS erroneously stores the return index on the second day as 10.152, which would result in a –90% return. These examples all show nonzero returns while the true return is zero. Additionally, we could have decimal errors when the true return

is nonzero. For example, when the return index decreases from 101.52 to 96.44 (–5% true return) but is stored as 964.40, this results in a computed return of 849.96%. This example shows that there is a need to account for decimal errors in both directions, whether the true return is zero or nonzero. We therefore omit any returns that are above 400% (a –50% true return accompanied by right-decimal error) or below –85% (a 50% true return accompanied by a left-decimal error). A second correction is to set both R_t and R_{t-1} to missing if R_t or R_{t-1} is greater than 300% and $(1 + R_{t-1})(1 + R_t) - 1$ is less than 50% (indicating extreme reversal), both for monthly and daily returns. Excess returns are calculated using the risk-free rate. MC and B/M are not corrected as no obvious errors are detected. In the analyses that follow, we use twelve lags for B/M to ensure that accounting data is always available to investors at the time.⁵ Since Jegadeesh and Titman (1993) the momentum effect is widely recognized and accepted in the literature on cross-sectional return predictability. Therefore, we also include it in our regressions and sorting algorithms. As in Fama and French (2008), momentum (*Mom*) in month t equals the buy-and-hold return of a particular stock over the period from month $t - 12$ to month $t - 2$. We use the return of month $t - 1$ (*LagRet*) separately to control for the short-term reversal effect documented by Jegadeesh (1990). We compute a very simple measure of expected illiquidity (*Illiq*) inspired by Bekaert et al. (2007). *Illiq* in month t is the proportion of zero returns observed over the last 260 trading days prior to month t . We did not use the Amihud (2002) measure as it would result in a major loss of observations: for the 2,971,458 firm-months, only 649,433 observations (or 22%) on volume are available. Lastly, we use the skewness coefficient (*Skew*) of the last 260 daily returns prior to month t as a proxy for expected skewness. If not all 260 daily returns are available, we ignore the missing values, but a minimum of 65 daily observations is required. If not, *Skew* is set to missing. We also calculate the four factors (market, *SMB*, *HML* and *WML*) advanced by Fama and French (1993) and Carhart (1997) to calculate portfolio alphas. The calculation of the factors is based on our own dataset. A detailed description is available in Appendix B.

As admitted by Bali et al. (2011), who estimate beta over a month using daily data, this estimate is subject to a significant amount of measurement error. We therefore decide to follow another, more robust approach. Market beta, *SMB* beta and *HML* beta in month t are calculated using a minimum of 24 and a maximum of 60 monthly excess stock returns from months prior to month t . If less than 24 returns are available, market beta is set to missing. The following time-series regression model is estimated⁶:

$$R_{i,t} - R_{f,t} = \alpha_i + b_i(R_{m,t} - R_{f,t}) + s_iSMB_t + h_iHML_t + \varepsilon_{i,t} \quad (1)$$

If the beta estimate is smaller than –1 or larger than 5, we set it to missing. These boundaries are quite arbitrary. However, the estimated betas are actually proxies of expected future beta and it would be hard to argue that any investor would expect more extreme betas. Not using these boundaries would result in betas as high as 50. We also need a proxy for expected idiosyncratic volatility (*IVol*) in month t . To account for the time-varying volatility of

³ The lists used are: WSCOPEOE, ALLAS, DEADOE (Austria); WSCOPEBG, FBDO, DEADBG (Belgium); WSCOPEFN, FFIN, DEADFN (Finland); WSCOPEFR, FFRA, ALLFF, DEADFR (France); WSCOPEBD, FGER1, FGER2, DEADBD1, DEADBD2 (Germany); WSCOPEIR, FIRL, DEADIR (Ireland); WSCOPEIT, FITA, DEADIT (Italy); WSCOPENL, FHOL, ALLFL, DEADNL (The Netherlands); WSCOPEPT, FPOM, FPOR, FPSM, DEADPT (Portugal); WSCOPEES, FSPN, DEADES (Spain); WSCOPELX, FLUX, DEADLX (Luxembourg); WSCOPEGR, FGREE, FGRPM, FGRMM, FNEXA, DEADGR (Greece); FSLOVAK, FSLOVALL, DEADSL (Slovakia).

⁴ This rate can be found on the website of the Deutsche Bundesbank in the time series database.

⁵ Using a constant lag of 12 months does not always reflect the most recent information. We may sometimes use the book-to-market ratio of 12 months ago when in fact new accounting data had already reached the market. We argue that the book value of equity is very persistent and that short-term variation in book-to-market ratios will mainly be caused by variation in stock prices. The fact that the value effect is still very strong in our data (see later) confirms this intuition.

⁶ We refrain from adding the WML factor as exposures with respect to it are much less persistent due to the relatively short-lived nature of momentum. Nevertheless, when we do use the WML exposure, differences in b , s and h across both factor models are minimal: correlations are respectively 0.93, 0.97 and 0.89. Moreover, the correlation between market betas from the three factor model and traditional CAPM betas is 0.88, or 0.80 when we use Dimson (1979) betas with one lag. Using these other estimators in our analysis does not impact the results in any significant way.

stock returns, *IVol* for month t is calculated as the root mean square error from a daily three factor model (where days are indexed by d) estimated on the 20 days prior to month t :

$$R_{i,d} - R_{f,d} = \alpha_i + b_i(R_{m,d} - R_{f,d}) + s_iSMB_d + h_iHML_d + \varepsilon_{i,d} \quad (2)$$

This is similar to the methodology of Ang et al. (2009).⁷ Lastly, we use minimum and maximum returns to measure extreme returns. More specifically, we use $MIN(N)$ or $MAX(N)$ to denote the average of respectively the N lowest or highest returns within a given estimation window. For example, $MIN(1)$ is the minimum return, while $MAX(3)$ is the average of the three highest returns. We use the absolute value of $MIN(N)$ for easier interpretation. The estimation window contains the 20 trading days prior to month t , to be sure that the measures are in the investors' information set. If there are more than five zero returns or missing values during these 20 days, MAX or MIN is set to missing, ensuring that MAX does not just capture a fairly low return from a series of zero returns. Sometimes we refer to MAX or MIN instead of $MAX(N)$ or $MIN(N)$ to avoid copious notation.

3. Results

In this section we discuss the results of our main analyses and robustness checks. We start with extensive sorts, we then move on to cross-sectional regressions at the firm level and end with the persistence of MAX .

3.1. Sorts

In panel A of Table 1, each month t , we sort stocks based on $MAX(N)$, with N ranging from 1 to 5. We then compute returns for the decile portfolios. Portfolio returns are either equal-weighted (EW) or value-weighted (VW). Each month, before sorting, the cross-section of returns is winsorized at the 0.5% and 99.5% levels to ensure that results are not driven by extremes, although the effect of winsorizing returns for portfolio sorts or regressions is found to be negligibly small. The table reports the time-series averages of EW and VW portfolio returns along with Newey and West (1987) adjusted t -statistics for the hedge portfolio, which is long in the high portfolio and short in the low portfolio. Four-factor alphas of the hedge portfolio and their Newey-West adjusted t -statistics are reported as *hedge 4F alpha*. We do not find significant indications of a negative MAX effect. Hedge returns are all negative, but show little statistical significance and are unimpressive, while alpha is only statistically significant for sorts on $MAX(1)$, with a t -statistic of -2.06 . All other EW or VW hedge returns or four-factor alphas are negative but cannot be significantly distinguished from zero. The effect weakens as the maxima are calculated using larger N . In any case, these results are much less impressive than those of Bali et al. (2011), who find strongly negative hedge returns and alphas for these sorts. Part of their more negative hedge returns can be explained by their low MAX portfolios, which seem to have higher returns than our low MAX portfolios. The pattern in their paper shows that deciles 1–9 have about the same average return, but then decile 10 shows a steep drop in average returns. The same holds in our sample, except that the low MAX decile also tends to have lower average returns (around 0.65%, compared to around 1.10% for deciles 2–9). However, when we look at the four-factor alphas of those low MAX deciles (not reported for brevity), we find no indication that they are different from alphas of deciles 2–9. Using Seemingly Unrelated Regression

(SUR), the hypothesis that the alphas of the first nine decile portfolios are equal is not rejected for EW $MAX(1)$ sorted portfolios ($p = 0.52$) or for their VW counterparts ($p = 0.81$). Additionally, the hypothesis that the alphas of all ten decile portfolios sorted on $MAX(1)$ are equal is not rejected ($p = 0.13$ for EW sorts and $p = 0.70$ for VW sorts). We conclude that the results from univariate sorts are weak in comparison with the results in Bali et al. (2011) and that they do not reveal a strong relationship between expected stock returns and MAX .

The insignificant hedge returns can potentially be explained by looking at panel B of Table 1. Here, each month t we compute the median of the cross-section of firm characteristics for each decile portfolios. The time-series averages of these medians are reported. We find that high MAX stocks have larger market betas, smaller MC and higher B/M : all of these are effects that have been found to increase average stock returns and therefore could distort or weaken the hedge returns found in panel A. On the other hand, high MAX stocks also tend to have lower momentum and higher lagged returns. These characteristics would bias the results towards lower returns. We also note that high MAX stocks exhibit lottery characteristics such as lower prices, higher idiosyncratic volatility and higher skewness. Lastly, low MAX stocks tend to have out-of-line skewness and illiquidity, compared to the monotonicity found for other variables. When we recompute panel B of Table 1 by only considering stocks in a certain month if their *Illiq* in that month is below 10%, these conspicuous characteristics diminish considerably. However, we stress that this liquidity constraint does not weaken the MAX effect, as will be shown later in the paper. In fact, using more liquid stocks often strengthens the effect.

In order to get a clearer picture of the MAX effect, we first resort to bivariate sorts that control for other effects. Afterwards, we turn to multivariate regressions. For Table 2, we form 10 portfolios by sorting stocks each month t on a control variable: market beta, MC , B/M , Mom or $LagRet$. Then we sort all stocks within each decile portfolio on $MAX(1)$ and form 10 portfolios within each control group so that 100 (10×10) portfolios are obtained, each with roughly an equal number of stocks. However, for brevity, we only report portfolio returns that are averaged across the spectrum of the control variable to produce decile portfolios with variation in MAX but with similar levels of the control variable, as do Bali et al. (2011). As such, we report the “average” MAX effect over all control deciles. Hedge returns are also averaged over the 10 portfolios, so that the hedge return is actually the return on a strategy that invests equal amounts of money in each hedge portfolio.

We find that the hedge returns are always negative. Moreover, they are statistically and economically significant for EW sorts on $Beta$ ($t = -1.96$), on B/M ($t = -2.54$) and on $LagRet$ ($t = -2.74$). For VW sorts, the hedge returns are not statistically significant. The hedge alphas are more convincing. They are all statistically and economically significant for all EW sorts and for all VW sorts except for those on B/M and $LagRet$. As with the univariate sorts, the results indicate that the first decile portfolio often has lower average returns than deciles 2–9. As before, this pattern largely disappears when we look at the four-factor alphas of the decile portfolios (not reported for brevity). We again use SUR to test the hypothesis that the alphas of the first nine decile portfolios are equal. The hypothesis can only be rejected for VW sorts controlling for momentum ($p < 0.001$) or lagged return ($p = 0.03$). The hypothesis that all ten alphas are the same, however, is often rejected: for EW sorts on $Beta$ ($p < 0.001$), B/M ($p < 0.001$), Mom ($p < 0.001$) and $LagRet$ ($p < 0.001$) and for VW sorts on $Beta$ ($p = 0.02$), Mom ($p < 0.001$) and $LagRet$ ($p = 0.01$). These results show that the alphas of the first nine deciles are usually similar, while the alpha of the high MAX decile is significantly lower. Lastly, in panel B we can see that sorting on $MAX(5)$ decreases the MAX effect, since all hedge returns and some alphas are now insignificant.

⁷ The correlation between our *IVol* measure and the regular standard deviation of the excess returns used to estimate *IVol* is 0.99. This shows that in the short run it does not really matter whether we use *IVol* or regular standard deviation.

Table 1
Univariate MAX sorts.

Portfolio	Average EW portfolio returns					Average VW portfolio returns					
	MAX(1)	MAX(2)	MAX(3)	MAX(4)	MAX(5)	MAX(1)	MAX(2)	MAX(3)	MAX(4)	MAX(5)	
Panel A: Average portfolio returns for univariate sorts on MAX(N)											
Low	0.69%	0.66%		0.63%	0.64%	0.62%	0.63%	0.63%	0.63%	0.65%	
2	0.92%	0.96%	0.95%	0.99%	0.97%	0.74%	0.73%	0.72%	0.72%	0.67%	
3	1.16%	1.16%	1.19%	1.18%	1.16%	1.02%	1.02%	0.97%	0.94%	0.95%	
4	1.32%	1.29%	1.28%	1.24%	1.27%	1.18%	1.16%	1.20%	1.16%	1.15%	
5	1.14%	1.18%	1.15%	1.19%	1.19%	1.00%	1.00%	0.97%	1.07%	1.14%	
6	1.09%	1.10%	1.14%	1.12%	1.13%	1.05%	1.12%	1.16%	1.07%	1.02%	
7	1.07%	1.12%	1.12%	1.14%	1.15%	0.99%	1.09%	1.09%	1.01%	0.95%	
8	1.35%	1.06%	1.00%	0.99%	1.01%	0.89%	0.90%	0.91%	0.94%	0.99%	
9	1.11%	1.23%	1.07%	1.03%	0.98%	0.92%	0.69%	0.66%	0.66%	0.67%	
High	0.22%	0.32%	0.52%	0.55%	0.57%	0.32%	0.36%	0.41%	0.47%	0.45%	
Hedge	−0.47%	−0.35%	−0.15%	−0.08%	−0.08%	−0.30%	−0.27%	−0.23%	−0.16%	−0.21%	
Hedge t	−1.20	−0.82	−0.30	−0.16	−0.15	−0.70	−0.62	−0.48	−0.35	−0.44	
Hedge 4F alpha	−0.55%	−0.47%	−0.30%	−0.26%	−0.26%	−0.48%	−0.46%	−0.45%	−0.40%	−0.48%	
Hedge 4F alpha t	−2.06	−1.68	−0.92	−0.76	−0.75	−1.34	−1.13	−1.06	−0.94	−1.16	
	MAX(1)	MIN(1)	Beta	MC (€10 ⁶)	B/M	Mom	LagRet	IVol	Illiq	Skew	Price (€)
Panel B: Averages of cross-sectional median of characteristics of decile portfolios sorted on MAX(1)											
Low	0.80%	1.04%	0.40	140.7	0.62	4.63%	−0.57%	0.44%	7.69%	0.45	56.27
2	1.49%	1.82%	0.63	201.3	0.65	8.28%	−0.96%	0.78%	5.93%	0.30	47.11
3	2.17%	2.41%	0.80	202.9	0.65	9.98%	−0.80%	1.06%	5.08%	0.28	41.79
4	2.85%	2.93%	0.94	193.5	0.65	10.96%	−0.45%	1.33%	4.86%	0.28	41.26
5	3.50%	3.33%	1.01	173.1	0.66	10.20%	0.01%	1.55%	4.92%	0.29	38.65
6	4.20%	3.69%	1.06	144.6	0.67	9.34%	0.67%	1.78%	5.15%	0.32	35.94
7	5.05%	4.13%	1.10	113.5	0.69	10.21%	1.26%	2.05%	5.51%	0.33	32.44
8	6.15%	4.69%	1.14	89.3	0.70	9.94%	2.48%	2.39%	5.86%	0.38	30.02
9	7.97%	5.40%	1.22	69.0	0.72	9.39%	4.14%	2.94%	6.65%	0.47	23.26
High	13.08%	7.34%	1.20	38.6	0.78	−3.26%	8.05%	4.57%	8.94%	0.89	16.13

Note: In panel A, each month t we sort stocks based on $MAX(N)$. We then form 10 portfolios so that each portfolio contains an equal number of stocks. Subsequent portfolio returns are either equally-weighted (EW) or value-weighted (VW). Each month, the cross-section of returns is winsorized at the 0.5% and 99.5% levels. The table reports the time-series averages of EW and VW portfolio returns, along with Newey and West (1987) adjusted t -statistics for the hedge portfolio, which is long in the high MAX portfolio and short in the low MAX portfolio. Four factor alphas of the hedge portfolio and their Newey and West (1987) adjusted t -statistics are reported as *Hedge 4F alpha*. In panel B, each month t we compute the median of the cross-section of firm characteristics for each decile portfolio. The time-series averages of these medians are reported. $MAX(N)$ is calculated by taking the average of the N highest returns in the estimation window, which contains the 20 daily returns prior to month t . $MIN(N)$ is calculated by taking the average of the absolute values of the N lowest returns in the estimation window. *Beta* is the market beta from the three factor model estimated using 24–60 monthly returns prior to month t . *MC* and *B/M* are the market capitalization and book-to-market ratio obtained from TDS. *Mom* is the buy-and-hold return from month $t - 12$ to month $t - 2$ and the *LagRet* is the return of month $t - 1$. *IVol* is the RMSE of the three factor model estimated using the 20 days prior to month t . *Illiq* is the proportion of zero returns observed in the 260 days prior to month t . *Skew* is the skewness of the 260 daily returns prior to month t . The sample period is 1980/01–2011/06.

The results of the bivariate sorts are more convincing than those of the univariate sorts. It is clear that controlling for other variables is very important to detect the MAX effect, which brings us to multivariate regressions.

3.2. Cross-sectional Fama–MacBeth regressions

Portfolio sorts are easy to interpret and they do not impose a functional form on the relationship between MAX and expected returns. However, a large amount of cross-sectional information is lost in the process of portfolio formation. Also, as the dependencies between different characteristics become increasingly complex, sorts may fail to provide a clear view on the relation between MAX and returns. Given the abundance of characteristics that seem to determine the cross-section of returns, we follow Fama and French (2008) and perform Fama and MacBeth (1973) regressions to disentangle the effects. Each month t we regress the cross-section of excess stock returns on k explanatory variables X_j ($j = 1, 2, \dots, k$):

$$R_i - R_f = b_0 + \sum_{j=1}^k b_j X_{ij} + \varepsilon_i \quad (3)$$

All explanatory variables are constructed so that investors know them prior to month t and are winsorized at the 0.5% and 99.5% levels. We only run a regression in month t if $n_t \geq 10(k + 1)$, where

n_t is the total number of observations in month t . We then focus on the time-series average of the estimates for b_j ($j = 1, 2, \dots, k$) to evaluate which characteristics are significantly related to the cross-section of stock returns. Statistical significance is based on Newey and West (1987) adjusted t -statistics.

Table 3 reports the results of these regressions using $MAX(1)$.⁸ We find that in a univariate regression (model 1) MAX is negatively yet insignificantly related to the cross-section of expected returns with an average slope coefficient of -0.0365 ($t = -1.72$). When we add the control variables (model 2), the MAX effect is stronger and significant with a coefficient of -0.0591 ($t = -3.63$). The other slopes are consistent with the literature: a positive but very small slope for beta ($t = 0.20$), a significantly negative size-effect ($t = -2.06$), a strong positive value effect ($t = 6.63$), a strong positive momentum effect ($t = 4.32$) and a significantly negative short term reversal effect ($t = -3.01$). The coefficient of *Illiq* is weakly significant ($t = -1.72$) but with a sign that contradicts our expectations: more liquid stocks have higher returns. Model 3 shows that dropping *Illiq* from the model does not significantly change the other coefficients much. Alternatively, we re-estimate model 2 by imposing an additional constraint. In a certain month, we only use stocks for which *Illiq* is

⁸ Returns and variables based on returns (e.g. *Mom*) are multiplied by 100 to express them as percentages. For example, the $MAX(1)$ coefficient in model 1 is -0.0365 . The interpretation is that an increase of 1 percentage point in $MAX(1)$ results in a -0.0365 percentage point decrease in excess returns, *ceteris paribus*.

Table 2
Bivariate MAX sorts.

Portfolio	Average EW portfolio returns					Average VW portfolio returns				
	Beta	Size	B/M	Mom	LagRet	Beta	Size	B/M	Mom	LagRet
<i>Panel A: bivariate sorts on control variables and MAX(1)</i>										
Low	0.88%	0.76%	0.95%	0.91%	1.09%	0.66%	0.78%	0.73%	0.67%	0.86%
2	1.11%	0.91%	1.12%	1.13%	1.14%	1.00%	0.90%	1.05%	0.96%	0.89%
3	1.17%	1.14%	1.33%	1.18%	1.11%	1.08%	1.13%	1.26%	0.95%	1.05%
4	1.19%	1.21%	1.28%	1.14%	1.27%	0.99%	1.18%	1.28%	1.01%	1.23%
5	1.09%	1.17%	1.14%	1.13%	1.14%	1.02%	1.13%	1.16%	0.96%	0.92%
6	1.13%	1.13%	1.04%	1.11%	1.14%	1.02%	1.10%	1.11%	0.77%	1.06%
7	1.13%	1.15%	1.16%	1.03%	1.10%	0.98%	1.14%	1.15%	0.78%	0.97%
8	1.19%	1.35%	1.22%	1.05%	0.95%	1.10%	1.30%	1.14%	0.68%	0.95%
9	1.26%	0.95%	1.14%	0.91%	0.82%	1.13%	0.90%	1.06%	0.80%	0.78%
High	0.36%	0.29%	0.20%	0.44%	0.28%	0.20%	0.22%	0.47%	0.20%	0.33%
Hedge	−0.52%	−0.47%	−0.74%	−0.47%	−0.81%	−0.47%	−0.56%	−0.26%	−0.46%	−0.53%
Hedge <i>t</i>	−1.96	−1.28	−2.54	−1.70	−2.74	−1.63	−1.53	−0.83	−1.78	−1.66
Hedge 4F alpha	−0.79%	−0.60%	−0.75%	−0.86%	−0.71%	−0.64%	−0.69%	−0.33%	−0.82%	−0.45%
Hedge 4F alpha <i>t</i>	−4.42	−2.41	−3.16	−4.48	−3.41	−2.68	−2.79	−1.17	−3.16	−1.77
<i>Panel B: summary of hedge portfolios for bivariate sorts on control variables and MAX(5)</i>										
Hedge	−0.24%	−0.19%	−0.47%	−0.40%	−0.75%	−0.33%	−0.29%	−0.30%	−0.45%	−0.57%
Hedge <i>t</i>	−0.58	−0.42	−1.18	−1.28	−2.26	−1.01	−0.66	−0.83	−1.60	−1.67
Hedge 4F alpha	−0.59%	−0.37%	−0.60%	−0.84%	−0.73%	−0.59%	−0.48%	−0.46%	−0.89%	−0.58%
Hedge 4F alpha <i>t</i>	−1.88	−1.22	−1.91	−3.94	−3.42	−2.08	−1.55	−1.40	−3.17	−1.98

Note: Each month *t* we first sort stocks based on a control variable: market beta (*Beta*), size (*MC*), book-to-market ratio (*B/M*), momentum (*Mom*) and lagged return (*LagRet*) and form 10 portfolios. We then sort all stocks within each portfolio on MAX(1). We form 10 portfolios within each control group so that the result is 100 (10 × 10) portfolios, each with roughly an equal number of stocks. However, for brevity, we only report portfolio returns that are averaged across the spectrum of the control variable to produce decile portfolios with variation in MAX but with similar levels of the control variable. Each month, the cross-section of returns is winsorised at the 0.5% and 99.5% levels. The table reports the time-series averages of EW and VW portfolio returns, along with Newey and West (1987) adjusted *t*-statistics for the hedge portfolio, which is long in the high MAX portfolio and short in the low MAX portfolio. Four factor alphas of the hedge portfolio and their Newey and West (1987) adjusted *t*-statistics are reported as *Hedge 4F alpha*. MAX(1) is the highest daily return over the 20 daily returns prior to month *t*. *Beta* is the market beta from the three factor model estimated using 24–60 monthly returns prior to month *t*. *MC* and *B/M* are the market capitalization and book-to-market ratio obtained from TDS. *Mom* is the buy-and-hold return from month *t* − 12 to month *t* − 2 and the *LagRet* is the return of month *t* − 1. The sample period is 1980/01–2011/06.

not higher than 10%, which is rather strict. As such, only the most liquid stocks are kept in the sample. This constraint causes the average number of firms used in monthly regressions to drop from 1331 to 913. The coefficient of MAX is −0.0708 and highly significant (*t* = −4.01), while the coefficient of *Illiq* is −5.4623 (*t* = −2.28). This negative and significant *Illiq* coefficient is a bit anomalous. Liquid stocks seem to earn higher returns. However, the effect is only significant in the sample of liquid stocks and not in the full sample. Selecting a sample based on *Illiq* will obviously introduce bias in the *Illiq* coefficient. Another explanation might be that the proportion of zero returns is actually a measure of liquidity, rather than illiquidity, as proposed by Mazza (2013).

Thus far we have found that there is a significant MAX effect and that it is robust to controls for intermediate term momentum, short-term reversal, and other control variables. Moreover, the MAX effect is economically important. For instance, moving from the 5th percentile to the 95th percentile of the MAX(1) distribution (where percentiles are averaged over all months) implies a reduction in average excess return of 0.73% per month according to model 2, keeping all other variables constant.

3.3. Persistence of MAX

If past lottery-like behavior does not lead to future lottery-like behavior, i.e. if MAX is not persistent, then there is no reason for investors to prefer high MAX stocks, especially given that they tend to have lower expected returns. Also, it would be incorrect to use past MAX as a proxy for expected future MAX. Therefore, we investigate the persistence of MAX by forming a month-to-month portfolio transition matrix, which shows the probability P_{ij} that a stock in decile *i* will be in decile *j* the next month. If MAX is not persistent (or random) we expect all probabilities to be approximately 10%. Table 4 shows the transition probabilities for MAX(1). We find that

the diagonal transition probabilities are always above 10%, especially for the outer deciles (low to low and high to high). Stocks in decile 10 (high MAX) have a 26.1% probability of remaining there next month. They even have a 52.9% probability of being in decile 8, 9 or 10 in the next month. Results are robust to other levels of *N*. In fact, the probability of high MAX(*N*) stocks to be in decile 8, 9 or 10 the next month increases to 55.5%, 57.0%, 57.6% and 57.4% for *N* set to respectively 2, 3, 4 and 5. Lastly, we also calculate transition probabilities for longer horizons. Stocks in the highest MAX decile have a 24.2% probability of being there in 2 months, 23.0% in 3 months, 21.0% in 6 months and 18.9% in 12 months. Similarly, they have a 49.6% probability of being in decile 8, 9 or 10 in 2 months, 47.9% in 3 months, 44.9% in 6 months and 41.6% in 12 months.

We also assess persistence by cross-sectionally regressing MAX onto its lag (not tabulated). We find a coefficient for lagged MAX of 0.36 (*t* = 36.23). Lastly, we examine the total distribution of stock returns for the high and low MAX deciles. Each month we place stock returns into one of 10 groups, based on their MAX(1). As we move through the sample, we keep adding returns to the 10 groups. At the end of this process we have 10 groups that each contain about 65,000 returns. Then we calculate descriptive statistics for the outer two groups, which contain next-month returns of stocks that were in the lowest or highest MAX at some point in time. However, we trim each group of returns at the 0.5% and 99.5% level to be sure that results are not driven by the most extreme observations. The EW average return of the low MAX group is 0.47%, while that of the high MAX group is −0.25%. The medians are respectively 0.12% and −1.20%. This shows once more that high MAX stocks underperform low MAX stocks. The high MAX group has a much higher standard deviation (17.72%) than the low MAX group (6.56%). Also, we find that the skewness of the high MAX group is 1.17, compared to 0.51 for the low MAX group. While

Table 3

Cross-sectional regressions of firm returns on firm characteristics.

Model/sample	MAX(1)	Beta	LogMC	LogB/M	Mom	LagRet	Illiq	Skew	MIN(1)	IVoLL
<i>Panel A: basic models on the full sample</i>										
Model 1	−0.0365 (−1.72)									
Model 2	−0.0591 (−3.63)	0.0265 (0.20)	−0.0764 (−2.06)	0.4437 (6.63)	0.0106 (4.32)	−0.0197 (−3.01)	−1.6653 (−1.72)			
Model 3	−0.0596 (−3.68)	0.0269 (0.20)	−0.0732 (−2.28)	0.4451 (6.62)	0.0107 (4.29)	−0.0191 (−2.87)				
<i>Panel B: euro and pre-euro samples</i>										
Model 4	−0.0856 (−3.99)	−0.1240 (−0.72)	−0.0460 (−0.73)	0.5056 (4.37)	0.0090 (1.88)	−0.0092 (−0.90)	−1.2469 (−0.71)			
Euro	−0.0395 (−1.76)	0.1378 (0.72)	−0.0988 (−2.25)	0.3979 (5.34)	0.0118 (5.29)	−0.0274 (−3.38)	−1.9745 (−2.24)			
<i>Panel C: post-1999 country samples</i>										
Model 6	−0.0925 (−2.61)	0.0399 (0.17)	0.0194 (0.34)	0.3447 (2.81)	0.0127 (2.19)	0.0046 (0.39)	1.3488 (1.33)			
Benelux	−0.1669 (−6.85)	−0.0369 (−0.23)	0.0929 (1.71)	0.2411 (2.18)	0.0148 (3.22)	−0.0229 (−2.88)	1.8857 (2.47)			
Model 7	−0.0882 (−4.74)	−0.1281 (−0.67)	0.0093 (0.17)	0.4637 (2.55)	0.0106 (2.02)	−0.0354 (−3.49)	−0.3576 (−0.51)			
Germany	−0.1407 (−6.26)	−0.1814 (−1.09)	0.0168 (0.26)	0.5030 (4.93)	0.0124 (2.40)	0.0133 (1.01)	0.1449 (0.08)			
Model 8	−0.0955 (−1.82)	−0.4503 (−2.19)	−0.1039 (−0.76)	0.5612 (4.42)	0.0045 (0.74)	−0.0102 (−0.77)	−5.6828 (−1.30)			
France	−0.0943 (−2.30)	0.4334 (1.54)	0.0275 (0.43)	0.4343 (2.10)	0.0078 (1.34)	0.0168 (1.26)	0.2059 (0.17)			
Model 9	−0.0569 (−3.47)	0.0257 (0.19)	−0.0805 (−2.18)	0.4469 (6.66)	0.0112 (4.54)	−0.0192 (−2.95)	−1.5752 (−1.69)	−0.0618 (−2.89)		
Italy	−0.0439 (−2.74)	0.0302 (0.23)	−0.0884 (−2.53)	0.4528 (6.89)	0.0114 (4.71)	−0.0214 (−3.29)	−1.5363 (−1.67)	−0.0674 (−3.19)	−0.0177 (−0.93)	−0.0246 (−0.72)
<i>Panel D: skewness, minimum returns and idiosyncratic volatility</i>										
Model 12										
Model 13										

Note: Each month t we regress the cross-section of excess stock returns onto several explanatory variables. Monthly regressions are skipped when there are not 10 observations per parameter estimate, e.g. 100 observations for a model with 1 constant and 9 explanatory variables. This procedure yields a time-series of slope coefficients. The table reports the time-series averages of these coefficients along with their Newey and West (1987) adjusted t -statistics in parentheses. The control variables are market beta ($Beta$), the log of market capitalization ($LogMC$), the log of the book-to-market ratio ($LogB/M$), momentum (Mom), lagged return ($LagRet$), illiquidity ($Illiq$), skewness ($Skew$), minimum return (MIN) and idiosyncratic volatility ($IVoLL$). Each month, all variables are winsorised at the 0.5% and 99.5% levels, except for models 6–11 where we winsorise at the 2.5% and 97.5% levels since those samples are much smaller. $MAX(1)$ is the highest daily return over the 20 daily returns prior to month t . $MIN(1)$ is the absolute value of the lowest daily return over the 20 daily returns prior to month t . $Beta$ is the market beta from the three factor model estimated using 24–60 monthly returns prior to month t . $LogMC$ and $LogB/M$ are the logs of market capitalization and book-to-market ratio obtained from TDS. Mom is the buy-and-hold return from month $t - 12$ to month $t - 2$ and the $LagRet$ is the return of month $t - 1$. $Illiq$ is the proportion of zero returns observed in the 260 days prior to month t . $Skew$ is the skewness coefficient of the 65–260 daily returns prior to month t . $IVoLL$ is the 1 month lagged RMSE of the three factor model estimated using the 20 days prior to month t . Excess returns and variables based on returns are multiplied by 100 to express them as percentages. The sample period is 1980/01–2011/06.

Table 4Transition matrix for $MAX(1)$.

	Low (%)	2 (%)	3 (%)	4 (%)	5 (%)	6 (%)	7 (%)	8 (%)	9 (%)	High (%)
Low	42.4	17.2	7.8	5.3	4.4	3.5	3.1	2.7	2.4	3.7
2	17.9	21.5	13.7	9.2	7.6	6.6	5.4	4.6	3.9	4.1
3	8.6	14.1	17.0	12.6	10.4	8.8	7.8	6.4	5.4	4.6
4	5.4	9.8	13.1	13.9	12.5	11.0	9.8	8.5	7.1	5.3
5	4.4	8.0	10.9	12.8	13.2	12.4	10.9	9.6	8.1	5.8
6	3.6	6.7	9.3	11.6	12.6	12.8	12.1	10.8	9.6	6.8
7	2.9	5.3	8.1	10.3	11.2	12.3	13.5	12.5	10.7	8.5
8	2.5	4.5	6.6	8.8	10.1	11.4	12.9	14.1	13.7	10.2
9	2.3	3.8	5.5	7.0	8.5	9.9	11.3	14.3	17.1	14.0
High	3.4	3.6	4.1	5.1	5.9	7.1	8.6	11.3	15.5	26.1

Note: Each month t we sort stocks based on $MAX(1)$. We then form 10 portfolios so that each portfolio contains an equal number of stocks. In month $t + 1$, we do the same. Then we check to which decile stocks have moved. The table shows average transition probabilities, giving us the probability that a stock in decile i (rows) will be in decile j (columns) the next month. $MAX(1)$ is the highest daily return over the 20 daily returns prior to month t . The sample period is 1980/01–2011/06.

the median is lower, all other percentiles above and including the 75th percentile lie much higher for the high MAX group. For example, the 75th percentile is 6.36% for the high MAX group, while 1.61% for the other. For the 90th percentile we find respectively

18.89% versus 6.69%, for the 99th percentile we find 63.22% versus 23.48% and lastly, for the maxima, we find 103.70% versus 41.70%. These results indicate that although high MAX returns underperform on average, their lottery-like characteristics (high skewness, high idiosyncratic volatility and high maximum returns) are persistent.

4. Robustness

It is possible that other explanations account for the MAX effect. In this section, we discuss several alternative explanations and conduct a battery of robustness checks. We discuss robustness with respect to measurement of MAX , with respect to large currency changes and country effects, with respect to skewness and with respect to idiosyncratic volatility.

4.1. Variations in MAX

It is of interest to look at some alternative specifications of MAX in the regressions. Replacing $MAX(1)$ by the average of the N highest returns in a 20 day interval, for N ranging from 2 to 5, does not change much. All coefficients are highly significant and negative. We also compute $MAX(1)$ using daily returns over the last 2, 3 and 6 months. If more than 25% of daily returns are zero or missing,

MAX is set to missing. We then use these estimates in model 2. Again, this does not change much. Coefficients are highly significant and negative at all horizons. Additionally, we estimate model 2 using lags of MAX(1) to test the persistence of the MAX effect. The effect is found to be negative and significant up to two lags. From three lags on, the effect is negative but cannot be statistically distinguished from zero. Lastly, we increase the lag of all explanatory variables by 1, 2, 3 and 6 months to investigate how a time lag impacts portfolio performance. The MAX coefficients are still significant and negative up to a 2 month's lag. These additional tests lead us to conclude that the MAX effect is robust to other specifications, but that it is short-lived.

4.2. Currency and country effects

In our sample, all returns in the pre-euro (before 1999) period are converted into euros. Therefore, MAX itself but also post-formation returns may be influenced by large currency changes, potentially distorting the results. Another concern is that the MAX effect is driven by a small fraction of countries in the sample. In this section, we will address these concerns.

A fast, easy and straight-forward way to check for currency effects is to focus on the post-1999 period, which is also the most recent and therefore perhaps the most relevant period. Results of this regression are reported in Table 3 under model 4. The MAX effect is negative and highly significant with a coefficient of -0.0856 ($t = -3.99$). Given that this coefficient is more negative than the one in the full sample, this means that the coefficient for the pre-euro period is closer to zero. Indeed, it is -0.0395 ($t = -1.76$). These simple tests show that the MAX effect prevails in the euro period and therefore is not the result of large currency changes. To address the concern that a small number of countries may be driving this effect, we estimate model 2 on the post-1999 samples of individual countries or groups of countries. These samples are not affected by currency changes and are largely free of bias from other countries. Since some countries are too small to analyze separately, they are grouped. First, we group Belgium, the Netherlands and Luxembourg (*Benelux*). Next, we analyze Germany, France and Italy separately, since their samples are large enough. We also group Spain, Portugal and Greece (*SPG*) together as they are commonly considered as peripheral European countries. The remaining countries, Austria, Finland, Ireland and Slovakia (*AFIS*), are grouped together.⁹ Models 6–11 show the results of regressions using the euro period for these samples. The t -statistics of the MAX coefficients all show clear evidence for a MAX effect in euro country subsamples. They are respectively -2.61 , -6.85 , -4.74 , -6.26 , -1.82 and -2.30 . Moreover, the coefficients are all economically significant. The MAX effect is particularly strong in the three largest samples (both in time-series and in the cross-section): Germany, France and Italy. It is remarkable that for each country or group of countries the coefficient of MAX(1) exceeds that in the pooled sample. A possible explanation is that MAX is not fully integrated across markets, since the investors who arguably drive the phenomenon are presumably underdiversified and possibly confine their holdings to their home country.

Lastly, another way to control for currency and country effects is to analyze returns in excess of their value-weighted home market return. The value-weighted market return for a particular country is computed by using all firms in that country with available data on market capitalization and returns. This procedure is a rough way to capture both currency changes and country effects in

post-formation portfolio performance. Some remarks are appropriate here. First, note that this is only a rough correction, since implicitly we assume a beta of one for all stocks with respect to their home market. If Brown and Warner (1980) provide any indication, this will not matter very much. Using more advanced models would lead us astray from our main objective, which is investigating the MAX effect. Also note that MAX is still measured in raw euro-denominated returns, since it would be hard to argue that investors measure lottery-like characteristics by looking at excess returns instead of raw returns. On the other hand one could argue that the ranking of MAX will not or barely change if currency or country effects are immaterial. Therefore, we test using both original MAX and MAX based on daily returns from which daily local market returns are subtracted. We denote this last variable as *adjusted MAX*.

We recalculate panel A of Table 1 and of Table 2 using this procedure. The results are reported in Table 5. In panels A and C we perform univariate sorts and in panels B and D we perform bivariate sorts. Portfolio performance is always based on country-adjusted returns, but we use original MAX in panels A and B and adjusted MAX in panels C and D. Comparing the results in panels A and B with Tables 1 and 2 immediately shows that all hedge returns and four-factor alphas have increased in size and significance. The results in panels C and D are even more impressive: hedge returns and alphas again increase in size and are more significant. For example, the adjusted EW hedge return for univariate sorts on adjusted MAX(1) is -0.79% ($t = -2.54$) with a corresponding alpha of -0.61% ($t = -2.75$). These sorts all confirm that the MAX effect is robust. Lastly, we run regression model 2 from Table 3 with country-adjusted returns and original MAX(1) (results not tabulated). We find a MAX coefficient of -0.0669 ($t = -4.92$) for the full sample and -0.0476 ($t = -2.73$) for the pre-euro sample. When we also use adjusted MAX(1), we find a coefficient of -0.0794 ($t = -6.22$) for the full sample and -0.0657 ($t = -3.83$) for the pre-euro sample. In all of these regressions, the other coefficients barely change.

We can conclude this section by stating that the MAX effect is robust to currency changes and that it is not the result of one or two countries driving the effect in the full sample. We have shown this by focusing on the euro period, by running the regressions separately for groups of countries and by adjusting returns and MAX for their local market return. In all of these cases, the MAX effect is unambiguously present and often even stronger.

4.3. Skewness

A growing body of literature has investigated the relationship between skewness and expected returns. Several asset pricing models that include a preference for skewness show that stocks with higher coskewness have lower expected returns, see e.g. Kraus and Litzenberger (1976) or Harvey and Siddique (2000). Moreover, several authors have argued that idiosyncratic skewness may also be priced. Simkowitz and Beedles (1978) and Conine and Tamarkin (1981) show that it can be optimal for investors to hold an underdiversified portfolio in order to capture return skewness. Following this insight, Mitton and Vorkink (2007) show that heterogeneity in skewness preference leads to a lack of diversification in equilibrium. They also show that total skewness, including idiosyncratic skewness, is important for asset pricing. Other papers that demonstrate the role of idiosyncratic skewness for asset pricing are for instance Barberis and Huang (2008), Brunnermeier and Parker (2005), Brunnermeier et al. (2007) and Boyer et al. (2010). If MAX is related to priced skewness measures and these measures are not included in the model, MAX could incorrectly be picking up the effect. In this section, we investigate this claim.

⁹ Since the samples are much smaller and winsorizing at 0.5% and 99.5% levels in e.g. a sample of 90 observations makes little sense, all variables are now winsorized at the 2.5% and 97.5% levels. Winsorizing at 0.5% and 99.5% levels, however, does not significantly impact results.

Table 5
Adjusted univariate and bivariate sorts.

Portfolio	Average EW portfolio returns					Average VW portfolio returns				
	MAX(1)	MAX(2)	MAX(3)	MAX(4)	MAX(5)	MAX(1)	MAX(2)	MAX(3)	MAX(4)	MAX(5)
<i>Panel A: univariate sorts on original MAX(N) – portfolio performance based on adjusted returns</i>										
Hedge	–0.59%	–0.51%	–0.37%	–0.32%	–0.32%	–0.36%	–0.39%	–0.34%	–0.31%	–0.37%
Hedge <i>t</i>	–1.62	–1.34	–0.89	–0.76	–0.74	–0.95	–0.98	–0.81	–0.75	–0.89
Hedge 4F alpha	–0.53%	–0.47%	–0.35%	–0.33%	–0.32%	–0.40%	–0.42%	–0.40%	–0.35%	–0.43%
Hedge 4F alpha <i>t</i>	–2.20	–1.96	–1.32	–1.18	–1.13	–1.22	–1.13	–1.03	–0.95	–1.17
	Beta	Size	B/M	Mom	LagRet	Beta	Size	B/M	Mom	LagRet
<i>Panel B: bivariate sorts on control variables and original MAX(1) – portfolio performance based on adjusted returns</i>										
Hedge	–0.72%	–0.60%	–0.92%	–0.60%	–0.90%	–0.59%	–0.69%	–0.42%	–0.58%	–0.56%
Hedge <i>t</i>	–3.18	–1.80	–3.53	–2.44	–3.22	–2.43	–2.07	–1.54	–2.65	–1.93
Hedge 4F alpha	–0.88%	–0.59%	–0.77%	–0.85%	–0.70%	–0.65%	–0.69%	–0.36%	–0.80%	–0.35%
Hedge 4F alpha <i>t</i>	–5.71	–2.72	–3.82	–5.01	–3.66	–3.13	–3.18	–1.44	–3.66	–1.58
	MAX(1)	MAX(2)	MAX(3)	MAX(4)	MAX(5)	MAX(1)	MAX(2)	MAX(3)	MAX(4)	MAX(5)
<i>Panel C: univariate sorts on adjusted MAX(N) – portfolio performance based on adjusted returns</i>										
Hedge	–0.79%	–0.67%	–0.65%	–0.62%	–0.60%	–0.64%	–0.85%	–0.80%	–0.80%	–0.70%
Hedge <i>t</i>	–2.54	–2.07	–1.87	–1.74	–1.64	–1.90	–2.42	–2.25	–2.12	–1.88
Hedge 4F alpha	–0.61%	–0.53%	–0.52%	–0.51%	–0.48%	–0.41%	–0.63%	–0.52%	–0.60%	–0.53%
Hedge 4F alpha <i>t</i>	–2.75	–2.18	–2.04	–1.93	–1.74	–1.33	–1.85	–1.49	–1.57	–1.40
	Beta	Size	B/M	Mom	LagRet	Beta	Size	B/M	Mom	LagRet
<i>Panel D: bivariate sorts on control variables and adjusted MAX(1) – portfolio performance based on adjusted returns</i>										
Hedge	–0.75%	–0.75%	–1.04%	–0.71%	–1.04%	–0.79%	–0.79%	–0.82%	–0.96%	–0.69%
Hedge <i>t</i>	–3.58	–2.61	–4.40	–3.40	–4.48	–3.93	–2.78	–3.37	–5.17	–2.71
Hedge 4F alpha	–0.79%	–0.73%	–0.82%	–0.83%	–0.80%	–0.72%	–0.75%	–0.61%	–1.05%	–0.47%
Hedge 4F alpha <i>t</i>	–4.75	–3.46	–4.52	–5.12	–4.81	–3.58	–3.53	–2.95	–5.27	–2.21

Note: In panels A and C, each month *t* we sort stocks based on respectively original MAX(*N*) and adjusted MAX(*N*). We then form 10 portfolios so that each portfolio contains an equal number of stocks. In panels B and D, each month *t* we first sort stocks based on a control variable: market beta (*Beta*), size (*MC*), book-to-market ratio (*B/M*), momentum (*Mom*) and lagged return (*LagRet*) and form 10 portfolios. We then sort all stocks within each portfolio on original MAX(1) for panel B and on adjusted MAX(1) for panel D. We form 10 portfolios within each control group so that the result is 100 (10 × 10) portfolios, each with roughly an equal number of stocks. However, for brevity, we only report portfolio returns that are averaged across the spectrum of the control variable to produce decile portfolios with variation in MAX but with similar levels of the control variable. Subsequent portfolio returns are either equally-weighted (EW) or value-weighted (VW). Portfolio performance is not based on returns as in Tables 1 and 2, but on adjusted returns. The adjusted return for a given stock (either daily or monthly) is the return of that stock reduced by the value-weighted market return of its home country. Original MAX(*N*) is calculated by taking the average of the *N* highest returns in the estimation window, which contains the 20 daily returns prior to month *t*, while adjusted MAX(*N*) is calculated in the same way but by using adjusted daily returns. Each month, the cross-section of adjusted returns is winsorised at the 0.5% and 99.5% levels. The table reports the time-series averages of EW and VW hedge portfolio returns, along with Newey and West (1987) adjusted *t*-statistics, where the hedge portfolio is long in the high MAX portfolio and short in the low MAX portfolio. Four factor alphas of the hedge portfolio and their Newey and West (1987) adjusted *t*-statistics are reported as Hedge 4F alpha. *Beta* is the market beta from the three factor model estimated using 24–60 monthly returns prior to month *t*. *MC* and *B/M* are the market capitalization and book-to-market ratio obtained from TDS. *Mom* is the buy-and-hold return from month *t* – 12 to month *t* – 2 and the *LagRet* is the return of month *t* – 1. The sample period is 1980/01–2011/06.

The calculation of total skewness (*Skew*) has already been discussed. To calculate idiosyncratic skewness and coskewness, we follow Harvey and Siddique (2000) and estimate the following model for each stock in each month, based on 65–260 prior daily returns:

$$R_{i,d} - R_{f,d} = \alpha_i + \beta_i(R_{m,d} - R_{f,d}) + \gamma_i(R_{m,d} - R_{f,d})^2 + \varepsilon_{i,d} \quad (4)$$

Coskewness is defined as the estimate of γ_i in the above model, while idiosyncratic skewness is defined as the skewness coefficient of the residuals $\varepsilon_{i,d}$. The average cross-sectional correlation between *Skew* and idiosyncratic skewness is 0.99, so we can confidently state that total and idiosyncratic skewness are identical for practical purposes.

Panel B of Table 1 shows that high MAX stocks tend to have higher total skewness. The average cross-sectional correlation between MAX(1) and *Skew* is 0.17 ($p < 0.001$). Also, the correlation between MAX(1) and idiosyncratic skewness is 0.16 ($p < 0.001$). In model 12 (Table 3) we added *Skew* to model 2 for the full sample. MAX and *Skew* are negatively priced with an average slope of respectively –0.0569 ($t = -3.47$) and –0.0618 ($t = -2.89$). Compared to model 2, the coefficient of MAX barely changed. The fact that total skewness is priced is interesting and its economic significance cannot be ignored: a *ceteris paribus* change of skewness

from its 5th percentile to the 95th percentile results in a decrease in monthly returns of 0.45%. When we use idiosyncratic skewness instead of *Skew*, we find an average slope of –0.0575 ($t = -2.80$) for idiosyncratic skewness and of –0.0571 ($t = -3.50$) for MAX.

The average cross-sectional correlation between MAX(1) and coskewness is only –0.03 ($p < 0.01$), so it is unlikely that MAX is a proxy for coskewness. However, briefly testing the impact of coskewness is interesting on its own. If we re-estimate model 12 with coskewness instead of *Skew*, we find an average slope of –1.1844 ($t = -2.81$) for coskewness and of –0.0579 ($t = -3.53$) for MAX. Adding both coskewness and idiosyncratic skewness to the model leads to a coefficient of –0.0559 ($t = -3.39$) for MAX, –1.2263 ($t = -2.86$) for coskewness and –0.0569 ($t = -2.75$) for idiosyncratic skewness. We conclude that the MAX effect is robust with respect to all skewness measures.

4.4. Idiosyncratic volatility

Lastly, we investigate the effect of idiosyncratic volatility. Ang et al. (2006, 2009) have recently documented the idiosyncratic volatility puzzle: there is a negative relationship between idiosyncratic volatility and future returns and this relationship holds in both US and international markets. Asset pricing theory would sug-

gest that idiosyncratic risk either does not command a risk premium (since it can be diversified away) or that it commands a positive risk premium (Merton, 1987). However, the negative premium that is found all over the world presents us with a puzzle. Ang et al. (2009) estimate Fama–MacBeth regressions for Germany, France and Italy, amongst other countries such as Canada, Japan and the UK, and find a significant negative relationship between idiosyncratic volatility and future US dollar denominated returns. Using a regression model with *Beta*, *LogMC*, *LogB/M*, *Mom*, *LagRet*, *Illi* and *IVol* on the Germany, France and Italy samples, we find parameter estimates (not tabulated) for *IVol* of respectively -0.2233 ($t = -4.57$), -0.0480 ($t = -1.01$) and -0.1970 ($t = -3.01$). These results are similar to the results in Table 2 of Ang et al. (2009).¹⁰ For the full sample we find an *IVol* coefficient of -0.0275 ($t = -0.75$), while in the post-1999 sample it is -0.0042 ($t = -0.11$). This is strikingly low and insignificant, however, we have reason to believe that this may be due to liquidity effects. A monthly elimination procedure that eliminates all stocks for which *Illi* is higher than 50% produces an *IVol* coefficient of -0.0990 ($t = -2.28$) for the full sample and -0.1780 ($t = -3.40$) for the post-1999 sample. Using an even more strict liquidity boundary of 10% does not change these results.

Maximum returns are closely linked to minimum returns and idiosyncratic volatility. Stocks with higher idiosyncratic volatility will by definition show higher minimum and maximum returns, and vice versa. The average cross-sectional correlation between *MAX*(1) and *IVol* in our sample is 0.87 ($p < 0.001$). Between *MIN*(1) and *IVol* it is 0.80 ($p < 0.001$) and between *MAX*(1) and *MIN*(1) it is 0.56 ($p < 0.001$). Does *MAX* capture the idiosyncratic volatility puzzle or is it a separate effect? It is not easy to assess this question. Bivariate sorting is not a good methodology, since the high correlation between the two variables will result in almost no variation in *MAX* within each decile of *IVol*, or vice versa. Sorting independently does not solve this problem, since the outer decile portfolios will contain very few or no stocks. Regression is a better option, but high correlation between explanatory variables may cause multicollinearity problems. Ahn et al. (2013) assessed the reliability of Fama–MacBeth estimators when factors loadings have high levels of cross-sectional correlation.¹¹ Their main conclusion is that when multicollinearity is high, the Fama–MacBeth two-pass estimation procedure can lead to seriously biased estimates, distorted finite-sample distributions of *t*-statistics and reduced power. Assessing multicollinearity can be done with the condition number (CN), which is popularly used in regression analysis. It is defined as the square root of the ratio of the maximum and minimum eigenvalues of the correlation matrix of explanatory variables. Ahn et al. (2013) show that CN's above 5 are very problematic for smaller samples (240 months or less) and CN's above 10 are problematic for larger samples. We calculate the CN each month by using the correlation matrix of all explanatory variables (*Beta*, *LogMC*, *LogB/M*, *Mom*, *LagRet*, *Illi*, *Skew*, *MAX*, *MIN* and *IVol*). We find an average CN of 7.66. The CN is larger than 5 for 95% of all months, larger than 8 for 35% of all months, larger than 9 for 22% of all months and larger than 10 for 11% of all months. This is evidence that multicollinearity may pose a

serious problem if we add *MAX* and *IVol* to the same regression model, and that a correction is necessary.

Nevertheless, we run the regression with all explanatory variables anyway (results not tabulated). Interestingly, we find a significant *MAX* coefficient of -0.0665 ($t = -2.54$) and a positive *IVol* coefficient of 0.0924 ($t = 0.96$). Using a *Illi* boundary of 10% results in a *MAX* coefficient of -0.0856 ($t = -3.01$) and an *IVol* coefficient of 0.1100 ($t = 0.95$). Double sorts confirm these results. When we first sort on *IVol* and then on *MAX*(1), the average EW return differential between high *MAX* and low *MAX* stocks is -0.36% ($t = -2.29$) with a corresponding four-factor alpha of -0.38% ($t = -2.47$). The VW return differential is -0.19% ($t = -1.03$) with a corresponding alpha of -0.17% ($t = -0.90$). First sorting on *MAX*(1) and then on *IVol*, however, produces an EW return differential between high *IVol* and low *IVol* stocks of 0.14% ($t = 0.75$) with a corresponding alpha of 0.13% ($t = 0.69$). The VW return differential is -0.27% ($t = -1.29$) with a corresponding alpha of -0.30% ($t = -1.39$). These results all suggest that the *MAX* effect is robust to adding *IVol* to the model. Moreover, they suggest that the idiosyncratic volatility puzzle is not robust after controlling for *MAX*, especially in EW portfolios. However, these results are based on regressions with strong multicollinearity or sorts with a high degree of correlation and are therefore not very reliable.

An easy way to reduce correlations (and therefore multicollinearity) is to lag *IVol* by 1 month, denoted *IVolL*. Lagging *IVol* is a reasonable solution for three reasons. First, the average cross-sectional correlation between *IVolL* and *MAX* is 0.44, which is a lot lower than 0.87. Second, it is better than lagging *MAX*. Lagging a variable in Fama–MacBeth regressions means that for that variable we replace the cross-section at time *t* by the cross-section at time *t* – 1. In this context, it is important that these cross-sections are not too different. We have already discussed that cross-sectional regressions of *MAX*(1) onto its lag produce an average slope of 0.36 ($t = 36.23$). Doing the same for *IVol* produces an average slope of 0.50 ($t = 62.89$). This means that *IVol* is more persistent than *MAX*, which explains why we choose to lag *IVol* to reduce multicollinearity. Third, the idiosyncratic volatility puzzle does not disappear by lagging *IVol*. In a model with *Beta*, *LogMC*, *LogB/M*, *Mom*, *LagRet* and *Illi* as control variables, the coefficient for *IVolL* is -0.0317 ($t = -1.02$) for the full sample and -0.0589 ($t = -1.54$) for the euro sample. Imposing the constraint that *Illi* has to be lower than 10%, which significantly changed coefficients using *IVol*, changes the *IVolL* coefficients to -0.0833 ($t = -1.60$) for the full sample and -0.1517 ($t = -1.54$) for the euro sample. Although statistical significance has decreased, the idiosyncratic risk puzzle remains. The coefficients are economically significant and lower statistical significance was to be expected from using a lagged version of *IVol*.

Model 13 in Table 3 shows the regression on the full sample including all explanatory variables (including *IVolL*). The average CN has dropped to 3.41 and is never higher than 5, which means that we have successfully mitigated the multicollinearity problem. The coefficient of *MAX* is -0.0439 ($t = -2.74$) and that of *IVolL* is -0.0246 ($t = -0.72$), while that of *MIN* is small and insignificant. Bivariate sorts lead to similar results. First sorting on *IVolL* and then on *MAX*(1) produces an average EW return differential between high *MAX* and low *MAX* portfolios of -0.49% ($t = -2.19$) with a corresponding four-factor alpha of -0.61% ($t = -3.13$). The VW return differential is -0.59% ($t = -2.30$) with a corresponding alpha of -0.54% ($t = -2.06$). However, first sorting on *MAX*(1) and then on *IVolL* produces an average EW return differential of -0.09% ($t = -0.38$) with a corresponding alpha of 0.06% ($t = 0.31$) and an average VW return differential of 0.04% ($t = 0.15$) with a corresponding alpha of 0.19% ($t = 0.76$). Estimating model 13 using only the stocks for which *Illi* is lower than 50% or 10% does not change the results. The post-1999 sample

¹⁰ The results for France are a bit off. However, their methodology is somewhat different. Although they also use Fama–MacBeth regressions, their sample ends in December 2003, they use three-factor regression slopes instead of firm characteristics, they define lagged return as the return over the previous 6 months, they do not use momentum or liquidity measures, their returns are in US dollars, and so on. If we re-estimate the regression on the French sample over 1980–2003 and using only stocks for which *Illi* is lower than 10%, we find a coefficient for *IVol* of -0.1259 ($t = -1.79$).

¹¹ Strictly speaking, we do not use factor loadings but firm characteristics. There is however no reason to suspect that this changes the conclusion of Ahn et al. (2013). Firm characteristics are often considered as proxies for factors loadings on risk factors, as e.g. Fama and French (1997) suggest.

also reveals a *MAX* effect, with or without liquidity constraints. *IVol* is not significant in any of these specifications. Taken together, these analyses suggest that the *MAX* effect is robust to controlling for idiosyncratic volatility.

5. Conclusions

Using an extensive and carefully filtered sample of euro area stocks we are able to corroborate the results of [Bali et al. \(2011\)](#) that stocks with extreme positive returns have systematically lower average returns. The result perseveres after wide-ranging robustness checks. Although stocks with extreme positive returns also display low momentum and have high lagged monthly returns, these characteristics do not explain their underperformance. This finding is consistent with individual investors preferring stocks with lottery characteristics that imply a low probability of earning a very high return. This preference may increase demand for these stocks, thereby lowering their expected returns. The fact that extreme positive return stocks are generally small, relatively illiquid high *B/M* stocks with high idiosyncratic return volatility is consistent with this interpretation. These are precisely the stocks where the influence of individual investors on pricing is arguably larger. We find that the extreme positive return effect is not the result of large currency changes or a small number of countries. The robustness checks also shed light on two related issues that are interesting in their own respect. First, we find unambiguous evidence that both idiosyncratic skewness and coskewness are negatively priced in the cross-section. Second, since [Ang et al. \(2006, 2009\)](#) reported a negative relation between idiosyncratic volatility and average returns, this puzzle has been central to empirical asset pricing. Initially we find corroborating evidence for this negative relation. However, when we combine idiosyncratic volatility and a measure of maximum returns, the relation between return and idiosyncratic volatility is less robust. However, neither idiosyncratic volatility nor the different skewness measures weaken the extreme positive return effect.

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Appendix A. Filtering procedure

A first filter makes sure that a stock is the “major” stock for a company, as TDS defines it, to eliminate cross-listings. Next, the currency has to be euro. A third filter checks the industry-type for each company. We do not want to include investments such as REITs, investment trusts, venture capital trusts, asset management funds, or other funds. Also, the expanded company name cannot contain “suspicious” words such as *pref*, *prf*, *%*, *duplicate*, *dupl*, *afv*, *vvpr* or *strip*. Next, we only allow the type of equity to be EQ (for equity). Another filter makes sure that the asset’s home country matched the chosen list’s country to avoid TDS errors. Also, the ISIN code has to start with the correct two letters (e.g. BE for Belgium or FI for Finland), or the ISIN code has to be missing. Lastly, we remove doubles from the list. These occur often since multiple lists per country are used. Doubles are removed based on their *type*, which is a unique identifier code in TDS.

Table B1

Descriptive statistics for the four factors.

	Market	SMB	HML	WML
Average return	0.56%	0.02%	0.76%	0.94%
t-Statistic	1.95	0.14	4.38	3.49
Minimum	−21.26%	−11.43%	−7.88%	−29.09%
25th percentile	−1.88%	−1.41%	−0.78%	−0.54%
Median	1.11%	0.17%	0.74%	1.35%
75th percentile	3.40%	1.72%	2.21%	3.39%
Maximum	14.35%	7.23%	12.63%	15.37%
Skewness	−0.69	−0.36	0.57	−1.41
Excess kurtosis	2.30	1.27	2.60	7.83

Note: Each month, the four factors are calculated as described in [Appendix B](#). The following table provides summary statistics for the four factors.

Table B2

Correlation matrix for the four factors.

	Market (excess)	SMB	HML	WML
Market (excess)	1.00	−0.43	0.16	−0.28
SMB	−0.40	1.00	−0.04	0.18
HML	0.16	0.00	1.00	−0.37
WML	−0.10	0.07	−0.32	1.00

Note: Each month, the four factors are calculated as described in [Appendix B](#). The following table provides a correlation matrix for the four factors. Numbers above the diagonal are Pearson product moment correlation coefficients, while numbers below the diagonal are Spearman rank correlation coefficients.

Appendix B. Construction of market, SMB, HML and WML factors

The market factor in month *t* is a value-weighted average constructed from the individual excess stock returns from month *t*. The *SMB* and *HML* factors are constructed using six value-weighted portfolios formed on *MC* and *B/M*. The portfolios, which are constructed at the end of each June and then held for a year, are the (independent) intersections of two portfolios formed on *MC* and three portfolios formed on *B/M*. The *MC* breakpoint for year *t* is the 80th percentile at the end of June of year *t*. [Fama and French \(1993\)](#) actually use the median of NYSE market capitalizations, but [Schmidt et al. \(2011\)](#) have shown for their European dataset (which is very similar to ours) that the 80th quantile is very similar to the size breakpoints used by [Fama and French \(1993\)](#). The *B/M* breakpoints are the 30th and 70th percentiles. The *SMB* factor is then the average return on the three low *MC* portfolios minus the average return on the three high *MC* portfolios. The *HML* factor is the average return on the two high *B/M* portfolios minus the average return on the two low *B/M* portfolios (the middle *B/M* group is not considered). The *WML* factor is constructed in a slightly different way. Six value-weighted portfolios are formed on *MC* and the prior *t* − 12 to *t* − 2 buy-and-hold return (*Mom*). The portfolios are constructed monthly and are the (independent) intersections of two portfolios formed on *MC* and three portfolios formed on momentum. The monthly size breakpoint is again the 80th percentile. The momentum breakpoints are the 30th and 70th percentiles. The *WML* factor is then the average return on the two high momentum portfolios minus the average return on the two low momentum portfolios. Descriptive statistics and correlations are provided in [Tables B1 and B2](#).

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